

# Nathnael Tesfaw

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## EDUCATION

### Cornell University, College of Engineering

Ithaca, NY

*B.S. in Computer Science, Minor in Artificial Intelligence (GPA: 3.5)*

*Aug 2023 – May 2027*

**Coursework:** Object-Oriented Programming & Data Structures, Computer Organization, Embedded Systems, Machine Learning, Cybersecurity, Analysis of Algorithms, Deep Learning

## EXPERIENCE

### Workday

Pleasanton, CA

*Software Engineer Intern (Incoming)*

*May 2026 – Aug 2026*

### Carnegie Mellon University

Pittsburgh, PA

*Software Engineering Intern*

*May 2025 – Aug 2025*

- Engineered a **GPT-4-powered cybersecurity extension** that proactively guided users toward secure behaviors, addressing human-error risks responsible for **~90%** of reported security incidents.
- Designed large-scale data processing pipelines in **Python (NumPy)** to analyze **6,000+ LLM outputs**, improving response actionability by **50%** through prompt iteration and scoring.
- Deployed a production backend on AWS EC2 via an **HTTPS Nginx** API gateway and DuckDNS **reverse proxy**, ensuring **end-to-end encryption**, data integrity, and reliable low-latency interactions across all requests.

### Rethink Food (Hack4Impact Cornell)

Ithaca, NY

*Lead Software Developer*

*Aug 2025 – Dec 2025*

- Led a **team of 7 developers and designers** to deliver a **production web application from requirements through deployment** for a nonprofit combating food insecurity across New York City.
- Built an interactive **data visualization platform** using **Mapbox** to display data from **10,000+ restaurants and community organizations**, improving transparency and donor engagement.
- Drove technical scoping and onboarding processes for **100+ applicants**, improving recruitment efficiency.

## PROJECTS

### DevAudit | TypeScript, Next.js, Node.js, PostgreSQL, Redis, Docker, AWS

*Jan 2026 – Apr 2026*

- Built an **autonomous AI security auditing agent** that clones GitHub repositories, runs **3 static analysis tools in parallel** (Semgrep, Bandit, Gitleaks), and streams live AI reasoning to the browser over WebSocket.
- Designed a **distributed pipeline** with BullMQ + Redis job queue, Express REST API, and a decoupled worker process; deployed via **Docker Compose on AWS EC2** with non-root containers and per-IP rate limiting.

### DreamBooth Re-implementation | PyTorch, Diffusers, HuggingFace

*Cornell CS 4782*

- Re-implemented **DreamBooth** (Ruiz et al. 2023) from scratch, fine-tuning **Stable Diffusion v1.5** for subject-driven image generation from as few as **3–5 reference photos**.
- Built **full UNet fine-tuning and LoRA variants** with prior-preservation loss, achieving CLIP-T prompt fidelity of **0.30**, matching the original paper's reported result of 0.305.
- Evaluated subject and prompt fidelity using **CLIP-I, DINO, and CLIP-T** metrics, reproducing the paper's finding that full fine-tuning outperforms LoRA on identity preservation.

### Greenzone (Hack4Impact Cornell) | Next.js, Node.js, Express.js, PostgreSQL, Git

*Jan 2025 – May 2025*

- Developed a **carrying capacity early-warning web app** to monitor rangeland degradation affecting **300,000+ herders** and **71 million livestock** across Mongolia.
- Implemented secure **RESTful APIs** and integrated Supabase to **ingest and visualize large-scale** remote-sensing and biomass datasets in real time, enabling data-driven land management decisions.

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript, C, Dart, Verilog, HTML/CSS, SQL, OCaml, RISC-V Assembly

**Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Flask, Flutter, NumPy, Pandas, Mapbox

**Databases & Cloud:** AWS (EC2), PostgreSQL, MongoDB, Firebase, Supabase, Google Cloud Platform, Microsoft Azure

**Tools:** Docker, Git, Nginx, DuckDNS, Heroku, Vercel, Linux